

Theorem (Cauchy–Schwarz Inequality). For any real numbers $a_1, \dots, a_n$ and $b_1, \dots, b_n$,

$$\left(\sum_{i=1}^n a_i b_i\right)^2 \leq \left(\sum_{i=1}^n a_i^2\right) \left(\sum_{i=1}^n b_i^2\right).$$

Equality holds iff the vectors $\mathbf{a} = (a_1, \dots, a_n)$ and $\mathbf{b} = (b_1, \dots, b_n)$ are linearly dependent; i.e. either one of them is the zero vector or there exists $\lambda \in \mathbb{R}$ such that $a_i = \lambda b_i$ for every i .

Proof. **1. Vector notation.** Set

$$\mathbf{a} = (a_1, \dots, a_n), \quad \mathbf{b} = (b_1, \dots, b_n) \in \mathbb{R}^n.$$

Their Euclidean inner product and norm are

$$\mathbf{a} \cdot \mathbf{b} = \sum_{i=1}^n a_i b_i, \quad \|\mathbf{a}\|^2 = \sum_{i=1}^n a_i^2, \quad \|\mathbf{b}\|^2 = \sum_{i=1}^n b_i^2.$$

Thus we must show $(\mathbf{a} \cdot \mathbf{b})^2 \leq \|\mathbf{a}\|^2 \|\mathbf{b}\|^2$.

2. A non-negative quadratic form. For $t \in \mathbb{R}$ define

$$Q(t) = \sum_{i=1}^n (a_i t - b_i)^2.$$

Each summand is a square, so $Q(t) \geq 0$ for every t .

3. Expand $Q(t)$.

$$\begin{aligned} Q(t) &= \sum_{i=1}^n (a_i^2 t^2 - 2a_i b_i t + b_i^2) \\ &= \left(\sum_{i=1}^n a_i^2\right) t^2 - 2\left(\sum_{i=1}^n a_i b_i\right) t + \sum_{i=1}^n b_i^2. \end{aligned}$$

Hence $Q(t) = At^2 + Bt + C$ with

$$A = \sum_{i=1}^n a_i^2, \quad B = -2 \sum_{i=1}^n a_i b_i, \quad C = \sum_{i=1}^n b_i^2.$$

4. Discriminant condition. A real quadratic polynomial that never becomes negative must have a non-positive discriminant, i.e. $B^2 - 4AC \leq 0$.

5. Compute the discriminant.

$$\begin{aligned} B^2 - 4AC &= \left(-2 \sum_{i=1}^n a_i b_i\right)^2 - 4 \left(\sum_{i=1}^n a_i^2\right) \left(\sum_{i=1}^n b_i^2\right) \\ &= 4 \left(\sum_{i=1}^n a_i b_i\right)^2 - 4 \left(\sum_{i=1}^n a_i^2\right) \left(\sum_{i=1}^n b_i^2\right). \end{aligned}$$

6. Derive the inequality. From $B^2 - 4AC \leq 0$ we obtain

$$4\left(\sum_{i=1}^n a_i b_i\right)^2 - 4\left(\sum_{i=1}^n a_i^2\right)\left(\sum_{i=1}^n b_i^2\right) \leq 0,$$

and dividing by the positive constant 4 yields

$$\left(\sum_{i=1}^n a_i b_i\right)^2 \leq \left(\sum_{i=1}^n a_i^2\right)\left(\sum_{i=1}^n b_i^2\right),$$

which is the Cauchy–Schwarz inequality.

7. Equality condition. Equality holds exactly when the discriminant is zero, i.e. when

$$\sum_{i=1}^n (a_i t - b_i)^2 = 0 \quad \text{for } t_0 = -\frac{B}{2A} = \frac{\sum_{i=1}^n a_i b_i}{\sum_{i=1}^n a_i^2}.$$

The sum of squares can vanish only if each term vanishes, so $a_i t_0 = b_i$ for all i . Hence either $\mathbf{a} = \mathbf{0}$ (trivial equality) or there exists $\lambda = t_0$ such that $a_i = \lambda b_i$ for every i . Thus equality occurs iff $\mathbf{a}$ and $\mathbf{b}$ are linearly dependent (including the degenerate zero-vector case).

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